NYC Airbnb Market Analysis (2019): Final Project Report

1. Project Overview

This project analyses the Airbnb New York City 2019 dataset to understand how the market is structured, what drives listing prices, and where demand is strongest. It follows a complete data analysis workflow: data understanding, cleaning, exploratory analysis, visualization, an interactive Power BI dashboard, and data-driven recommendations. The analysis was done in a Kaggle notebook (Python, pandas, Plotly), and the dashboard was built in Power BI. Only the dataset provided with the task was used.

Questions addressed

- What are the most common listing types?
- Which areas have the highest activity?
- How do prices differ between areas?
- How do price, reviews and other variables relate to each other?
- What factors appear to affect prices?
- Which areas show the strongest performance?
- Are there clear patterns or trends?

2. Dataset Information

- **Source:** AB_NYC_2019.csv (Airbnb New York City 2019).
- **Size:** 48,895 rows and 16 columns (7 integer, 3 float and 6 text columns).

Column	**Meaning**
id, host_id	Listing and host identifiers
name, host_name	Listing title and host name
neighbourhood_group, neighbourhood	Borough and neighbourhood
latitude, longitude	Listing coordinates
room_type	Entire home/apt, Private room or Shared room
price	Nightly price in USD
minimum_nights	Minimum stay required
number_of_reviews, reviews_per_month, last_review	Review count, monthly review rate and date of the last review
calculated_host_listings_count	Number of listings the host has
availability_365	Days per year the listing is available

Important limitation: the dataset has no rating column. The number of reviews and reviews per month are therefore used as a proxy for booking activity, not for guest satisfaction.

3. Data Cleaning Process

Issue	Action	Result
Missing name (16) and host_name (21)	Filled with “Unknown”, then both columns dropped	No impact on analysis
Missing reviews_per_month (10,052)	All belonged to listings with 0 reviews, so filled with 0	Logically correct
Missing last_review	Left empty to avoid inventing dates	10,043 remain after cleaning
last_review stored as text	Converted to datetime	Time analysis possible
Duplicates	Checked on full rows and on id; none found	No action needed
Price = 0	11 rows removed	Invalid records
minimum_nights > 365	14 rows removed	Unrealistic values
Price outliers (> $500)	Flagged with is_price_outlier, not removed (2.1% of listings)	Luxury listings kept
Stray spaces in text	Stripped	Consistent text
New features	is_inactive, has_reviews, host_type, stay_type, price_category, last_review_year, log_price	Ready for analysis

The cleaned dataset has **48,870 rows and 22 columns** (25 rows removed, 0.05%) and was saved as AB_NYC_2019_clean.csv. The IQR rule (upper bound $334) was judged too strict for New York, so a $500 threshold was used to flag outliers instead.

4. Exploratory Analysis

Price distribution. Price is strongly right-skewed (median $106, mean about $153, skewness 19.1). A log transform makes it nearly symmetric (skewness 0.63), which is why it was used for distribution and correlation work.

Listing types. Entire homes make up 52.0% of listings, private rooms 45.7% and shared rooms 2.4%.

Activity by area. Manhattan (21,654 listings, 44.3%) and Brooklyn (20,089, 41.1%) hold 85.4% of the market, followed by Queens (11.6%), the Bronx (2.2%) and Staten Island (0.8%). The largest neighbourhoods are Williamsburg (3,917 listings), Bedford-Stuyvesant (3,709) and Harlem (2,656).

Prices by area. For listings up to $500, the median price is $145 in Manhattan, $90 in Brooklyn, $75 in Queens and Staten Island, and $65 in the Bronx. In every borough, 75% of listings cost $200 or less.

What affects price. Price has weak correlations with all numeric variables ($|r| \leq 0.11$). Location and room type matter far more: the median entire-home price is $191 in Manhattan, $145 in Brooklyn, $120 in Queens and $100 in the Bronx and Staten Island, and Manhattan is the most expensive in all three room types.

Price, reviews and demand. Reviews per month depend mostly on minimum stay: the median falls from 1.00 (1-3 nights) to 0.40 (4-7), 0.26 (8-30) and 0.23 (31+), and the correlation between minimum nights and reviews per month is -0.25. Cheaper room types are reviewed slightly more often (shared 1.47, private 1.44, entire home 1.31 reviews per month on average). Listings above $500 are much less likely to have any review (60% versus 82% in the $76-150 group).

Strongest performance. Median reviews per month (reviewed listings) is highest in Staten Island (1.41), the Bronx (1.38) and Queens (1.21), and lowest in Brooklyn (0.66) and Manhattan (0.61). Manhattan holds 44.3% of listings but only 40.0% of reviews.

Hosts and inactive listings. 35.9% of listings have zero available days. Single-listing hosts are 86.2% of hosts and own 66.1% of listings, but 46.2% of their listings are inactive (median availability 6 days). Hosts with 6 or more listings are about 1% of hosts and own 9.8% of listings, with median availability of 283-319 days.

Time pattern. The last_review column holds only each listing's most recent review, so it shows recency of activity, not growth. 65% of last-review dates fall in 2019, with a sharp rise in June 2019 (12,596 listings) because the data is a snapshot taken in July 2019 (latest date: 8 July 2019).

5. Important Visualizations

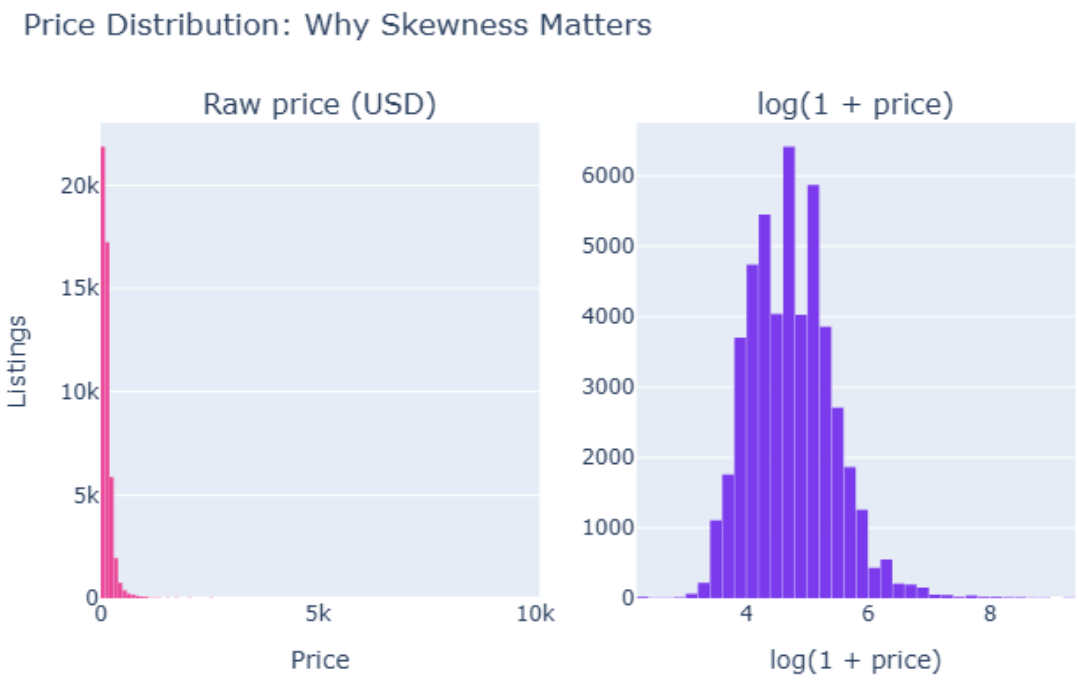


Figure 1. Price is heavily right-skewed; the log scale reveals the typical range.

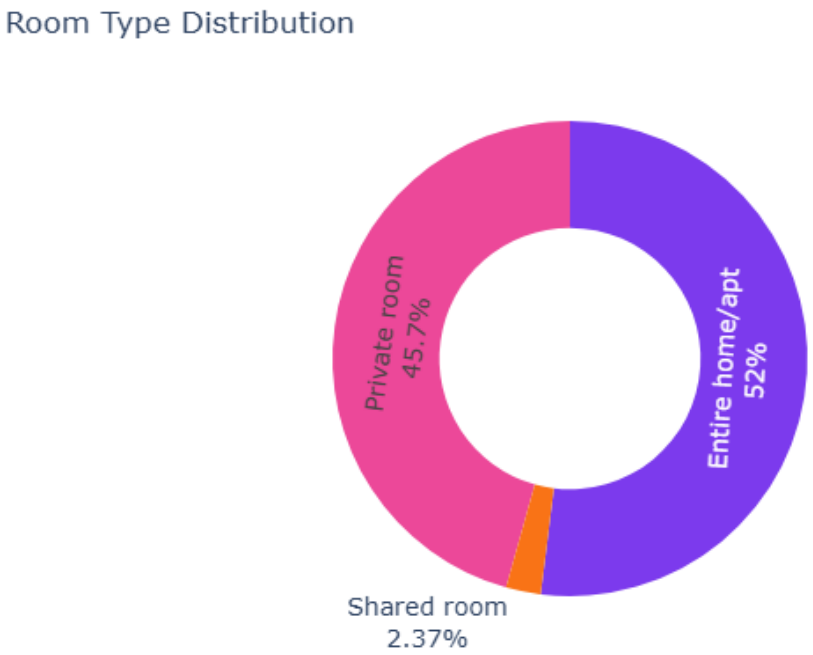


Figure 2. Entire homes and private rooms account for about 98% of listings.

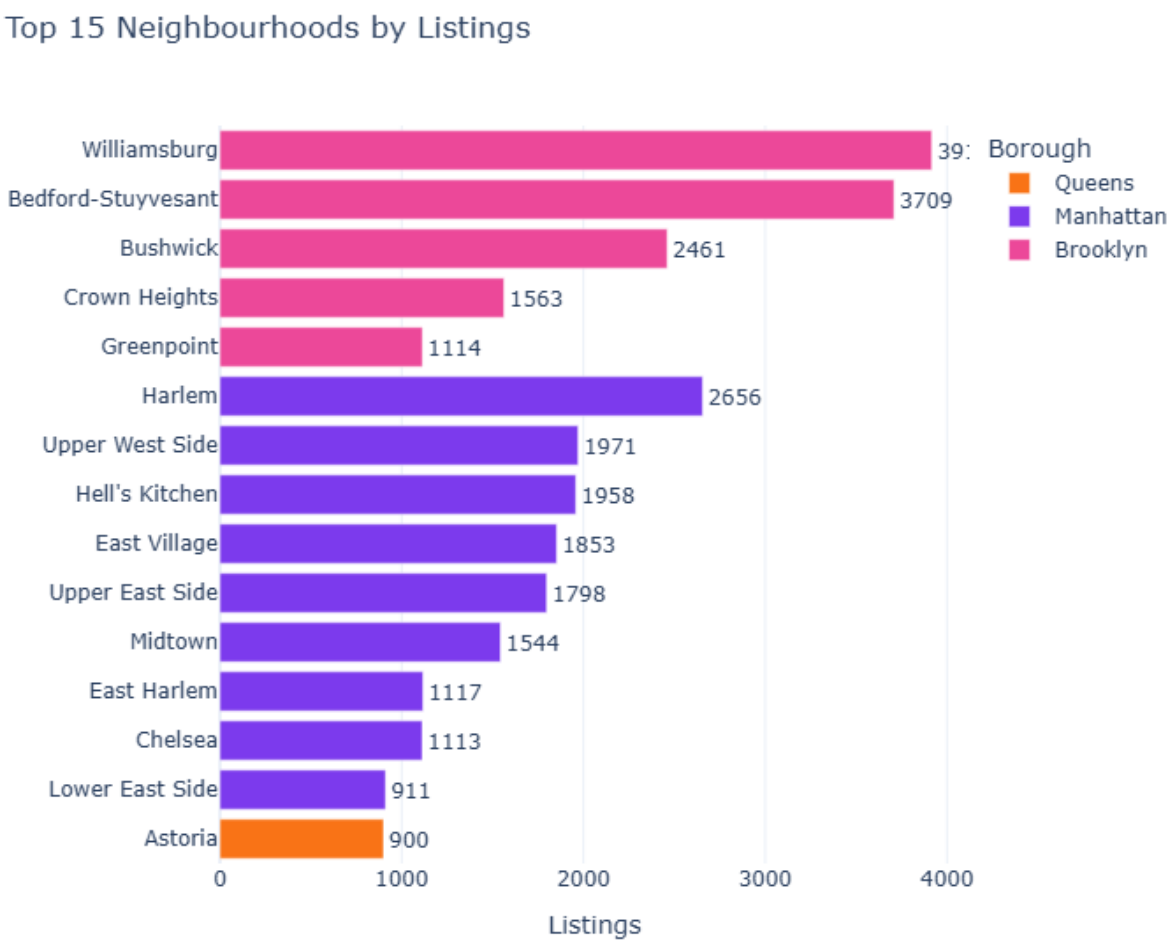


Figure 3. Williamsburg, Bedford-Stuyvesant and Harlem are the most active neighbourhoods.

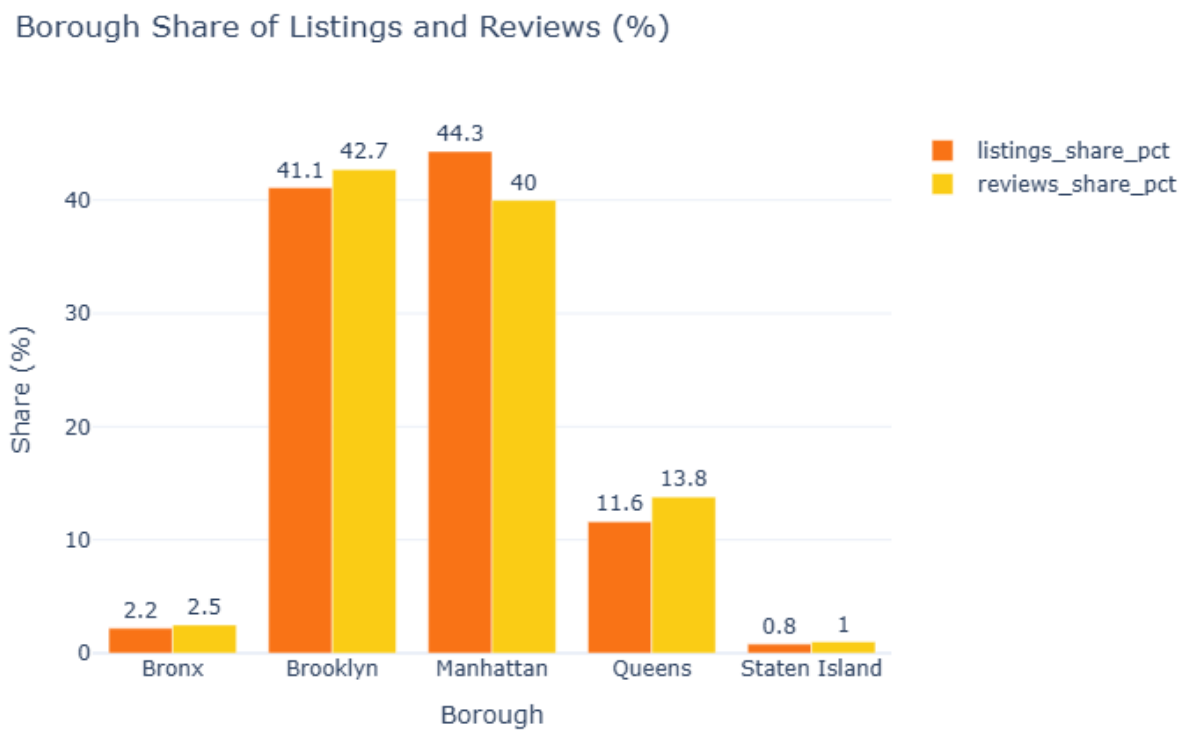


Figure 4. Manhattan's share of reviews is lower than its share of listings; Queens shows the opposite.

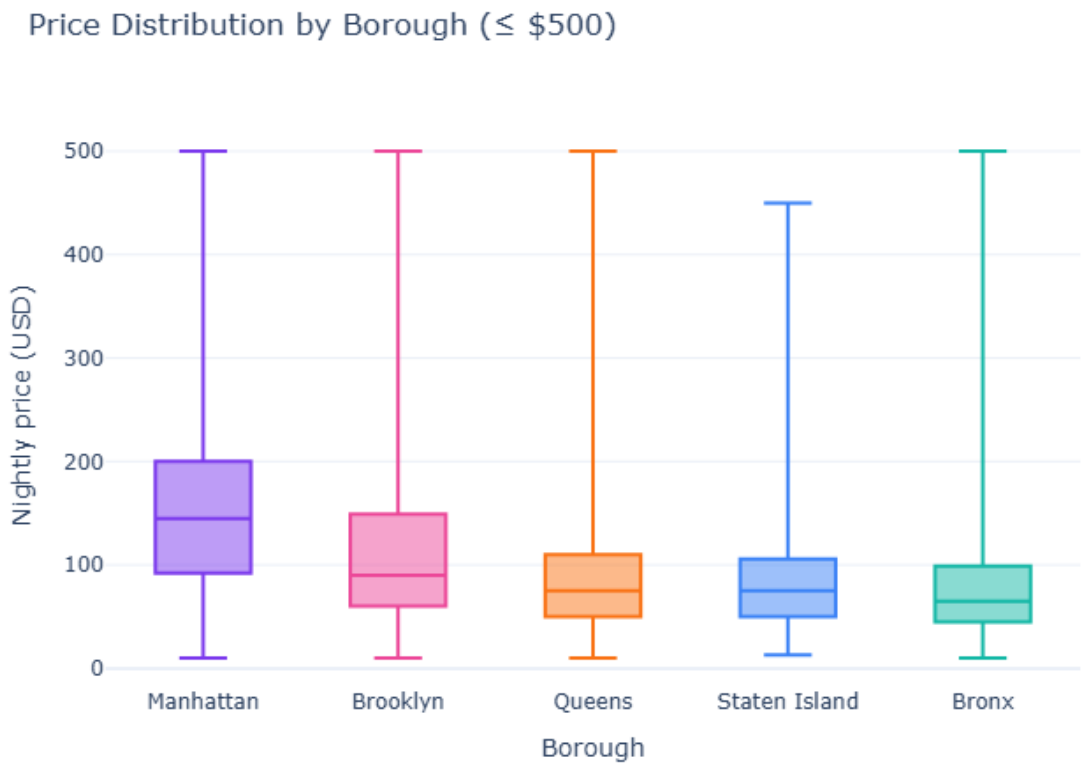


Figure 5. Manhattan has the highest and widest price distribution.

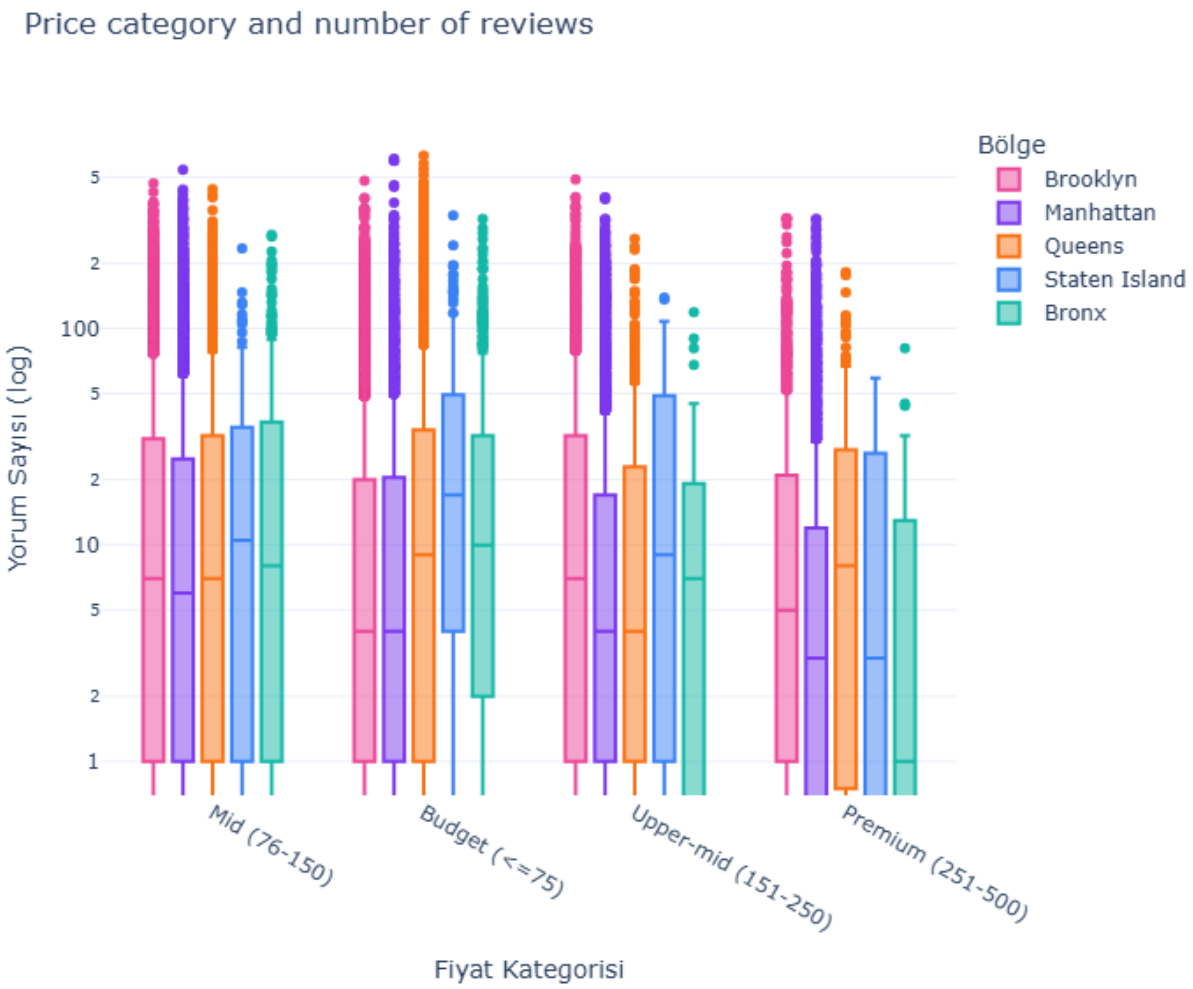


Figure 6. Review counts drop in the luxury price category.

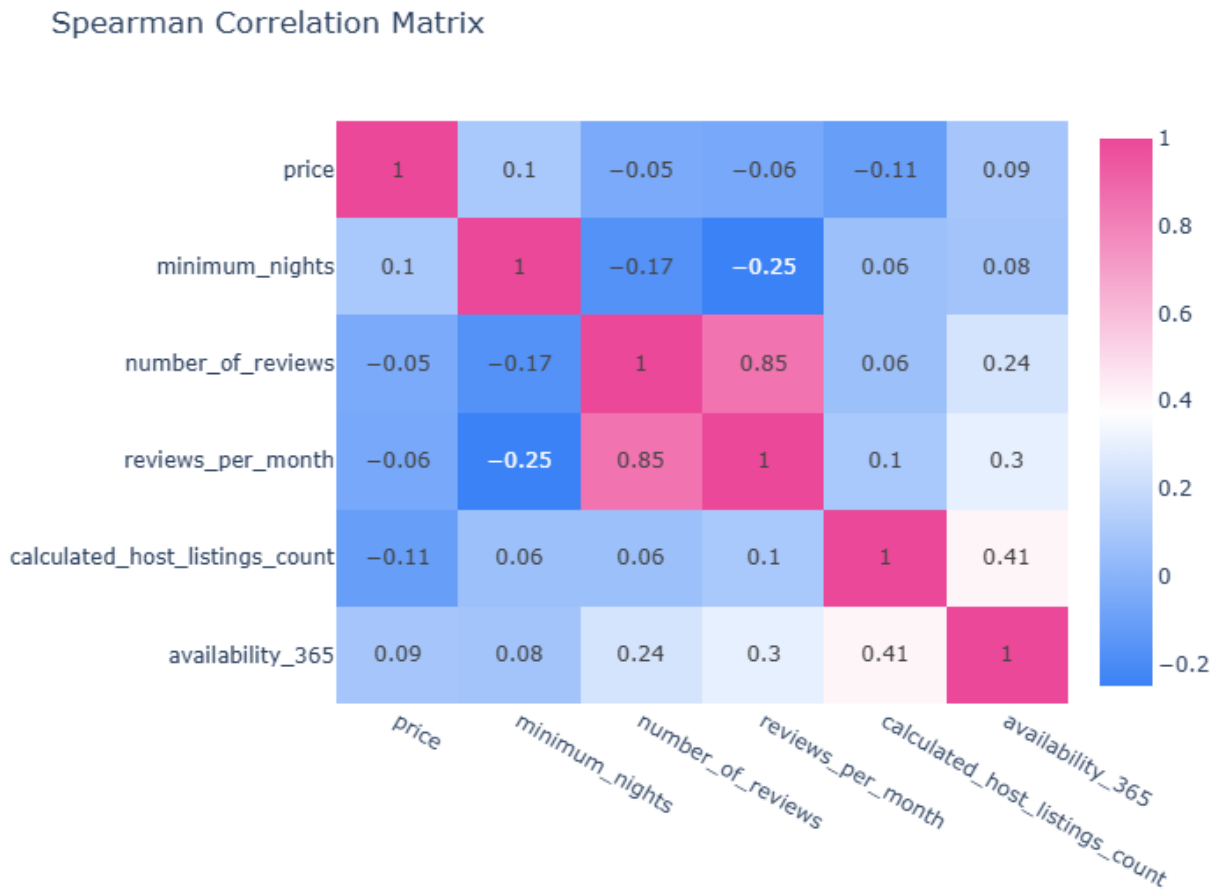


Figure 7. Price has only weak correlations with numeric variables; minimum nights and reviews per month are negatively related.

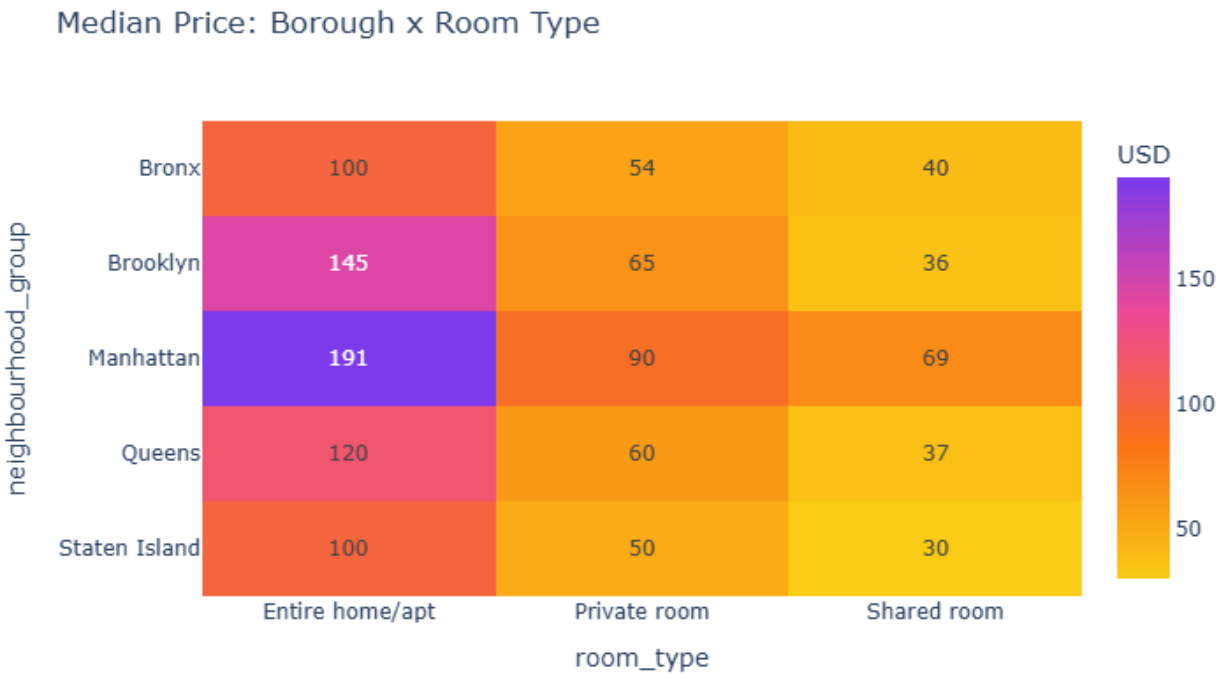


Figure 8. Manhattan is the most expensive in every room type, so the premium is a real location effect.

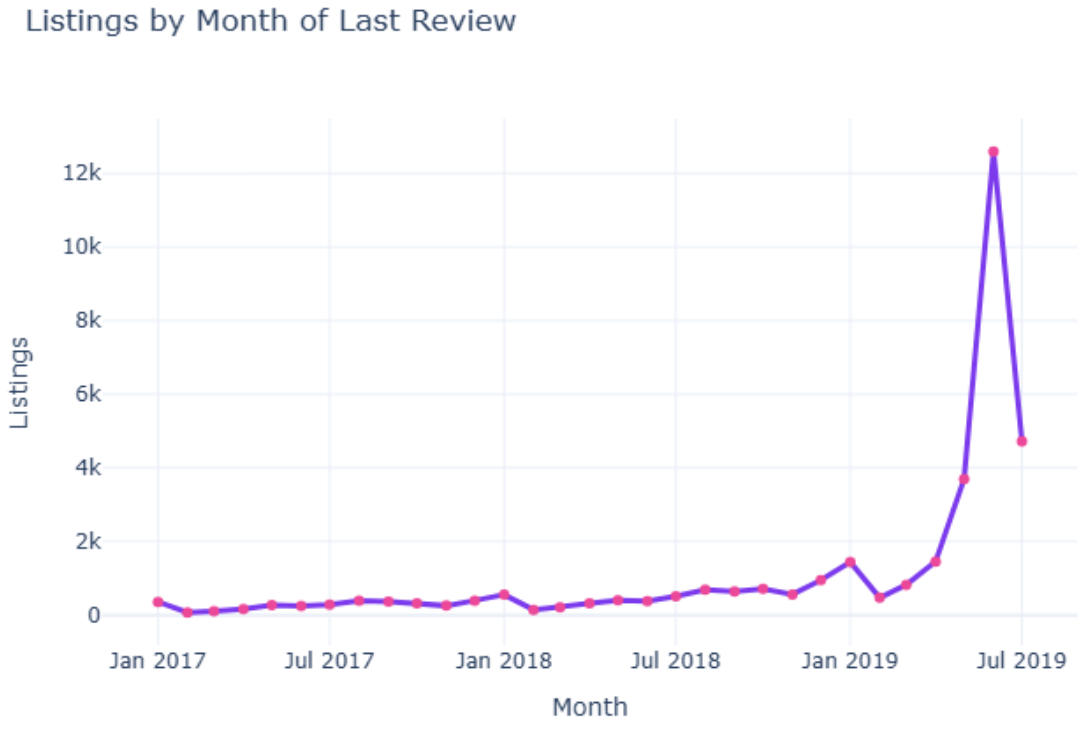


Figure 9. Most last-review dates cluster in mid-2019 because of the snapshot date.

6. Dashboard

The Power BI dashboard has three pages with a shared header (title, page navigation) and a slide-out filter panel with four slicers: borough, room type, price category and host type. The filters apply across all pages, and every chart cross-filters the others.

Overview covers supply and location. It has five KPI cards (total listings 48,870; median price $106; entire-home share 52.0%; multi-listing share 33.9%; average availability 113 days), a map of listings by borough, a room-type donut, listings by borough, and the top 15 neighbourhoods.

Pricing & Demand covers price and demand. It has four KPI cards (median price $106; reviews per listing 23.28; median reviews per month 0.72; reviewed listings share 79.4%), price quartiles by borough, a borough x room type price heatmap, median reviews per month by minimum stay, a neighbourhood performance bubble chart (neighbourhoods with at least 50 reviewed listings), and the last-review trend.

Key Insight summarises the main insights and recommendations of the project.

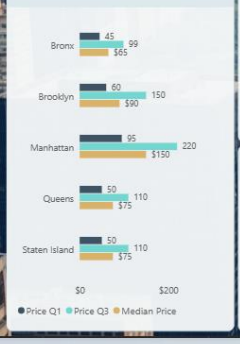
The notebook uses a box plot for price by borough; the dashboard shows the same quartiles (Q1, median, Q3) as columns because a box plot requires a custom visual.



NYC AIRBNB MARKET DASHBOARD

OVERVIEW | Pricing & Demand | Key Insight

Median Price	Reviews per Listing	Median Reviews per Month	Reviewed Listings Share
$106	23.28	0.72	79.4%







Median Price: Borough x Room Type

neighbourhood_group	Entire home/apt	Private room	Shared room	Total
Bronx	$100	$54	$40	$65
Brooklyn	$145	$65	$36	$90
Manhattan	$191	$90	$69	$150
Queens	$120	$60	$37	$75
Staten Island	$100	$50	$30	$75
Total	**$160**	**$70**	**$45**	**$106**



NYC AIRBNB MARKET DASHBOARD

OVERVIEW | Pricing & Demand | Key Insight

Business Insights & Recommendations

Insight 1: The market is concentrated in two boroughs and two room types

- Manhattan (**44.3%**) and Brooklyn (**41.1%**) hold **85.4%** of all listings. Entire homes (**52.0%**) and private rooms (**45.7%**) make up about **98%** of supply.
- Competition is intense in a small part of the city, and shared rooms (**2.4%**) are a negligible niche.
- Any pricing or growth decision is effectively a decision about Manhattan and Brooklyn entire homes and private rooms.
- Benchmark against listings in the same borough and room type, not against the citywide average.

Insight 2: Location and room type drive price, not popularity

- The median entire-home price is **$191** in Manhattan, **$145** in Brooklyn, **$120** in Queens and **$100** in the Bronx and Staten Island. Price has weak correlations with all numeric variables (**|r| ≤ 0.11**).
- The Manhattan premium exists within every room type, so it is a real location premium and not only a mix effect.
- Reviews or availability say little about what a listing can charge.
- Set prices from a borough x room type benchmark (the heatmap in the dashboard), then adjust for the listing's own features.

Insight 3: Demand per listing is higher outside Manhattan and Brooklyn

- Median reviews per month (reviewed listings): Staten Island **1.41**, Bronx **1.38**, Queens **1.21**, Brooklyn **0.66**, Manhattan **0.61**. Manhattan holds **44.3%** of listings but only **40.0%** of reviews.
- Core boroughs look more saturated, while outer boroughs get more booking activity per listing. Reviews measure booking activity, not satisfaction.
- New supply may earn more bookings where competition is lighter.
- Consider Queens first for new listings (largest outer-borough base, **5,664** listings). Treat the Bronx and Staten Island as promising but small samples (**1,090** and **373** listings).

Insight 4: Demand drops in the luxury segment

- Only **2.1%** of listings cost more than **$500**. Among them, **60%** have at least one review versus **82%** in the **$76-150** group, and the median review count is **6** versus **10**.
- Very high prices attract far fewer bookings.
- Overpricing leaves listings unbooked.
- Keep standard listings in the **$76-150** range, and price above **$500** only with a clear differentiator such as location or exceptional features.

Insight 5: Shorter minimum stays are booked more often

- Median reviews per month fall from **1.00** (**1-3** nights) to **0.40** (**4-7**), **0.26** (**8-30**) and **0.23** (**31+**). The correlation between minimum nights and reviews per month is **-0.25**.
- Longer minimums mean fewer guests per month, hence fewer bookings.
- Minimum nights is a setting the host controls directly.
- Hosts who want more bookings should test a minimum stay of **1-3** nights. This is an association, not proof of causation.

Insight 6: A third of listings are not bookable, and professional hosts dominate the open calendars

- **35.9%** of listings have zero available days (**46%** among single-listing hosts). Hosts with **6+** listings are about **1%** of hosts but hold **9.8%** of listings, with median availability of **283-319** days versus **6** days for single-listing hosts.
- Visible supply overstates what guests can actually book, and bookable supply is shaped by multi-listing operators.
- Market size and competition are overestimated if inactive listings are counted.
- Filter to active listings (availability > **0**) when benchmarking or sizing the market, and compare individual hosts against active, comparable listings.

Limitations

- The dataset has no rating column; reviews are used as a proxy for booking activity.
- The data is a single snapshot (latest review date: **July 2019**), so last_review shows recency of activity, not seasonality or growth.
- Results are associations; the dataset cannot prove causes.

7. Key Findings

1. Manhattan and Brooklyn hold 85.4% of listings, and entire homes and private rooms hold about 98%.
2. Location and room type drive price; popularity measures barely relate to price.
3. Median reviews per month are about twice as high in Queens, the Bronx and Staten Island as in Manhattan and Brooklyn.
4. Only 2.1% of listings cost more than $500, and they are much less likely to be reviewed.
5. Listings with 1-3 night minimums receive about 1.0 reviews per month versus 0.23-0.40 for longer minimums.
6. 35.9% of listings have a blocked calendar, while hosts with 6+ listings keep calendars open almost all year.

8. Business Insights

Insight 1: The market is concentrated in two boroughs and two room types.
What it means: competition is intense in a small part of the city, and shared rooms are a negligible niche. *Why it matters:* citywide averages are poor benchmarks, so any pricing or growth decision is effectively about Manhattan and Brooklyn entire homes and private rooms.

Insight 2: Location and room type drive price, not popularity.
What it means: the Manhattan premium exists inside every room type (entire home $191 versus $100 in the Bronx), so it is a genuine location premium and not only a mix effect. *Why it matters:* reviews and availability say little about what a listing can charge.

Insight 3: Demand per listing is higher outside Manhattan and Brooklyn.
What it means: core boroughs look more saturated, while outer boroughs record more booking activity per listing. Reviews measure activity, not satisfaction. *Why it matters:* new supply may earn more bookings where competition is lighter.

Insight 4: Demand drops in the luxury segment.
What it means: very high prices attract far fewer bookings (60% of listings above $500 have a review versus 82% in the $76-150 group; median reviews 6 versus 10). *Why it matters:* overpricing leaves listings unbooked.

Insight 5: Shorter minimum stays are booked more often.
What it means: longer minimums mean fewer guests per month and therefore fewer bookings. This is an association, not proof of causation. *Why it matters:* minimum nights is a setting the host controls directly.

Insight 6: A third of listings are not bookable, and professional hosts dominate open calendars.
What it means: visible supply overstates what guests can actually book, and bookable supply is shaped by multi-listing operators. *Why it matters:* market size and competition are overestimated if inactive listings are counted.

9. Recommendations

1. **Price by benchmark.** Set prices from a borough x room type benchmark (Insight 1 and 2), then adjust for the listing's own features.
2. **Explore outer boroughs.** Consider Queens first for new listings, as it has the largest outer-borough base (5,664 listings) and 1.21 median reviews per month. Treat the Bronx and Staten Island as promising but small samples (1,090 and 373 listings).
3. **Keep standard listings in the $76-150 range.** Price above $500 only with a clear differentiator such as location or exceptional features (Insight 4).
4. **Test shorter minimum stays.** Hosts who want more bookings should try a 1-3 night minimum and compare results (Insight 5).
5. **Benchmark using active listings only.** Filter to listings with availability above zero when sizing the market or comparing against competitors (Insight 6).

10. Conclusion

The New York City Airbnb market is highly concentrated, and prices are set mainly by where a listing is and what type it is, not by its popularity. Booking activity is higher outside the core boroughs and for cheaper, shorter-stay listings, and a large share of listed supply is not actually bookable. Together these findings suggest that hosts and investors should benchmark by borough and room type, look beyond Manhattan and Brooklyn, avoid very high prices without a differentiator, and consider shorter minimum stays.

Limitations. The dataset has no rating column, so reviews serve only as a proxy for booking activity. The data is a single snapshot from July 2019, so last_review reflects recency rather than seasonality or growth. The Bronx and Staten Island have small samples. All results are associations; the data cannot prove causes.